

## Spring 2025 - SOLUTION

1. a) Consider a schema **R(A,B,C,D,E,F,G)** with the following set of functional dependencies.

$A \rightarrow C$   
 $AB \rightarrow DE$   
 $CF \rightarrow D$   
 $C \rightarrow GFD$   
 $D \rightarrow F$   
 $G \rightarrow A$

3  
 +3  
 +2  
 =8

- i) Find out all possible **candidate keys**.  
 ii) Compute the **minimal cover** of the set of functional dependencies.  
 iii) Based on the **minimal cover**, examine each functional dependency and **determine** whether the relation is in **2NF** or not.

### Solution:

#### 1)a) i)

here,  $\alpha = \{B\}$ ,  $\beta = \{E\}$

Attribute closure,  $(B)^+ = \{B\}$  ; Not a super key

$(AB)^+ = \{A,B,C,D,E,F,G\}$  ; Super key

$(BC)^+ = \{A,B,C,D,E,F,G\}$  ; Super key

$(BD)^+ = \{B,D,F\}$  ; Not Super key

$(BF)^+ = \{B,F\}$  ; Not super key

$(BG)^+ = \{A,B,C,D,E,F,G\}$  ; Super key

So, candidate key =  $\{AB, BC, BG\}$

#### 1)a) ii)

| Decomposition                                                                                                                                                                                | Trivial remove                                                                                                                                                                              | Extraneous attribute removal                                                                                                                                           | Extraneous Functional dependency removal                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $A \rightarrow C$<br>$AB \rightarrow D$<br>$AB \rightarrow E$<br>$CF \rightarrow D$<br>$C \rightarrow G$<br>$C \rightarrow F$<br>$C \rightarrow D$<br>$D \rightarrow F$<br>$G \rightarrow A$ | $A \rightarrow C$<br>$AB \rightarrow D$<br>$AB \rightarrow E$<br>$C \rightarrow D$<br>$C \rightarrow G$<br>$C \rightarrow F$<br>$C \rightarrow D$<br>$D \rightarrow F$<br>$G \rightarrow A$ | $A \rightarrow C$<br>$AB \rightarrow D$<br>$AB \rightarrow E$<br>$C \rightarrow G$<br>$C \rightarrow F$<br>$C \rightarrow D$<br>$D \rightarrow F$<br>$G \rightarrow A$ | $A \rightarrow C$ ; $(A)^+ = \{A\}$ ; keep<br>$AB \rightarrow D$ ; $(AB)^+ = \{A,B, E\}$ ; Keep<br>$AB \rightarrow E$ ; $(AB)^+ = \{A,B,D,F\}$ ; Keep<br>$C \rightarrow G$ ; $(C)^+ = \{C, D, F\}$ ; Keep<br>$C \rightarrow D$ ; $(C)^+ = \{C,G,A\}$ ; Keep<br>$D \rightarrow F$ ; $(D)^+ = \{D\}$ ; Keep<br>$G \rightarrow A$ ; $(G)^+ = \{G\}$ ; Keep |
|                                                                                                                                                                                              | No trivial found                                                                                                                                                                            | $CF \rightarrow D$ is extraneous, according to three extraneous finding formulas.<br>Rest is fine.                                                                     | <b>So, Minimal cover is =</b><br>$\{A \rightarrow C, AB \rightarrow D, AB \rightarrow E, C \rightarrow G, C \rightarrow D, D \rightarrow F, G \rightarrow A\}$                                                                                                                                                                                          |

#### 1)a) iii)

From,  $i \rightarrow$  candidate keys are =  $\{AB, BC, BG\}$

Prime attributes (appear in at least one candidate key) :  $\{A, B, C, G\}$

Non-prime attributes:  $\{D, E, F\}$

Minimal cover is =  $\{A \rightarrow C, AB \rightarrow D, AB \rightarrow E, C \rightarrow G, C \rightarrow D, D \rightarrow F, G \rightarrow A\}$

Now, we have to check all FD from minimal cover.

$A \rightarrow C$ ;  $A \subset AB$  (candidate key) and  $C$  is prime  $\rightarrow$  no violation

$AB \rightarrow D$ ;  $AB$  is a full candidate key and  $D$  is non-prime  $\rightarrow$  no violation

### REMINDER

A relation is in Second Normal Form (2NF) if:

1. It is in 1NF, and
2. No partial dependency: A non-prime attribute (i.e. not part of any candidate key) is dependent on part of a composite primary key, but not on the whole key.

$AB \rightarrow E$ ; AB is full candidate key and E is non-prime  $\rightarrow$  no violation

$C \rightarrow G$ ; C is a part of candidate key BC, and G is prime  $\rightarrow$  no violation

**$C \rightarrow D$ ; C is a part of candidate key BC, but D is non-prime attribute and it violates the 2NF**

$D \rightarrow F$ ; D is not in candidate key, and both D, F is non-prime attribute.  $\rightarrow$  no violation

$G \rightarrow A$ ; G is a part of Candidate key BG, and A is prime attribute.  $\rightarrow$  no violation

**So, the relation R is not in 2NF, as Functional Dependency  $C \rightarrow D$  violates the 2NF.**

#### Extra: check whether the relation is in 3NF:

Candidate Keys: { AB, BC, BG }

Prime attributes: {A, B, C, G}

Non-prime attributes: {D, E, F}

Minimal cover is = { $A \rightarrow C$ ,  $AB \rightarrow D$ ,  $AB \rightarrow E$ ,  $C \rightarrow G$ ,  $C \rightarrow D$ ,  $D \rightarrow F$ ,  $G \rightarrow A$ }

**Let's check the all FD from Minimal cover:**

$A \rightarrow C$ ; A is not a super key; C is prime  $\rightarrow$  no violation

$AB \rightarrow D$ ; AB is candidate key  $\rightarrow$  no violation

$AB \rightarrow E$ ; AB is candidate key  $\rightarrow$  no violation

$C \rightarrow G$ ; G is a part of Candidate key BG  $\rightarrow$  no violation

**$C \rightarrow D$ ; C is not a superkey and D is not a prime attribute or part of a candidate key  $\rightarrow$  violation**

**$D \rightarrow F$ ; D is not a superkey and F is non-prime attribute or part of a candidate key  $\rightarrow$  violation**

$G \rightarrow A$ ; A is a part of candidate key.  $\rightarrow$  no violation

**So, the relation R is not in 3NF, as Functional Dependency  $C \rightarrow D$ ,  $D \rightarrow F$  violates the 3NF.**

#### REMINDER

For every functional dependency:  $X \rightarrow Y$

The relation is in 3NF if at least one of the following holds:

1. X is a super key, OR
2. Y is a prime attribute (part of some candidate key)

If neither is true  $\rightarrow$  3NF violation

b) Suppose we have a relation named **R**, having attributes as **P, Q, R, S, T** and **U**. The relation **R** is having the following functional dependencies,

$QR \rightarrow S$

$S \rightarrow T$

$TU \rightarrow P$

$P \rightarrow Q$

i) Considering the relation with functional dependencies, find if  $PR \rightarrow T$  is valid or not using **Armstrong's axioms**. [mention the axioms applied in each state]

ii) If we decompose the relation in **R1(P, Q)** and **R2(S, T)**, **R3(P, R, S, U)** then **determine** whether this decomposition is **lossless** or **lossy**.

#### 1)b) i)

We have to find the attribute closure of PR using given FD's, If it provides T we will say it's valid.

Attribute closure  $(PR)^+ = \{P, R, Q, S, T\}$ ; So, given functional dependency  $PR \rightarrow T$  is valid.

#### 1)b) ii)

Given, Decomposed Relations

R1(P, Q)

R2(S, T)

R3(P, R, S, U)

**Step 1: Identify common attributes**

$R1 \cap R2 = \emptyset \rightarrow$  no common attributes  $\rightarrow$  no need to check.

$$R1 \cap R3 = \{P\}$$

$$R2 \cap R3 = \{S\}$$

### Step 2: lossless decomposition test

$$R1 \cap R3 = \{P\}; \text{ Is } P \text{ a key for } R1(P, Q)?$$

Yes, P determines Q, so P is a key for R1

$$R2 \cap R3 = \{S\}; \text{ Is } S \text{ a key for } R2(S, T)?$$

Yes, S determines T, so S is a key for R2

Since each intersection is a key for one of the corresponding relations, then the decomposition is lossless.

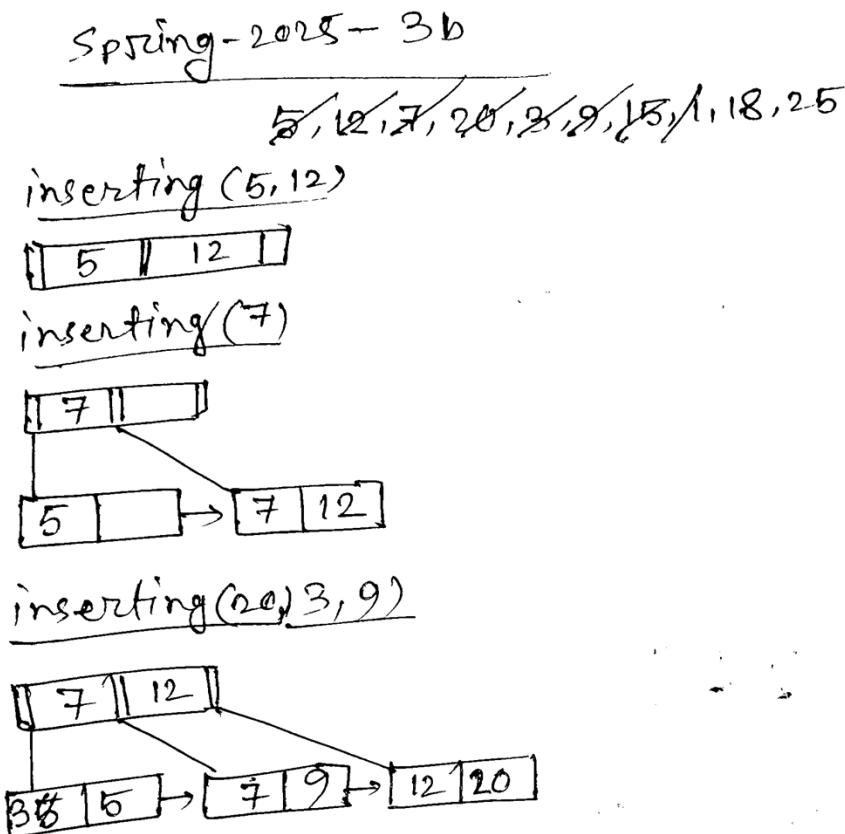
*Note: If the intersection attributes is a key (determinant) in any of the table then it's lossless, otherwise loosely.*

3. a) Clearly explain the difference between indexing and hashing in databases. In what scenarios is each method preferable? 3

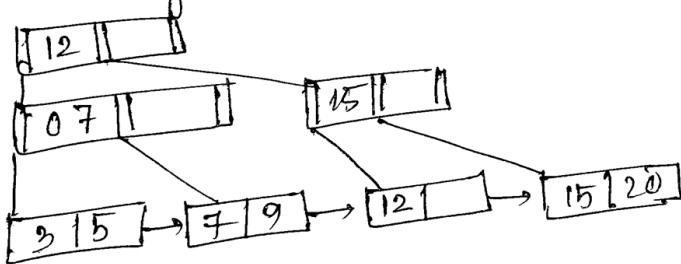
b) Consider the following data coming from left to right. Construct a **B+ Tree of order 3** for the incoming data and show all necessary steps while inserting. 7

5, 12, 7, 20, 3, 9, 15, 1, 18, 25

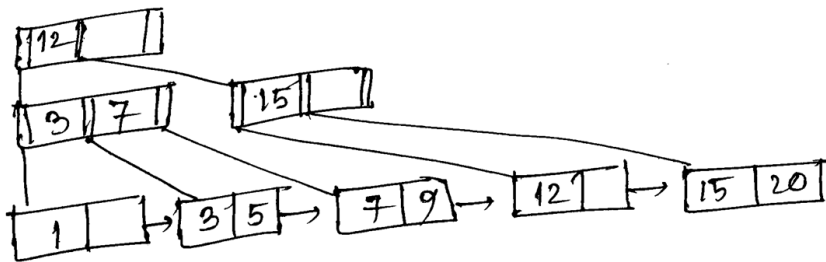
3)b)



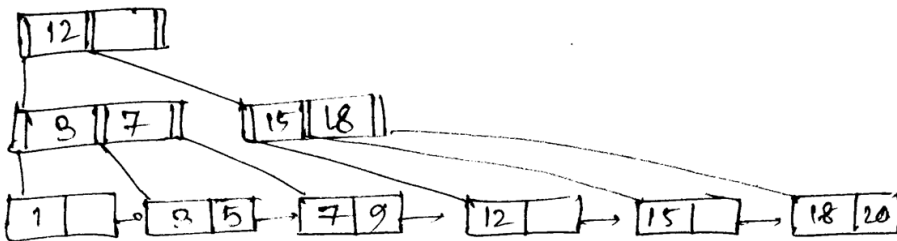
inserting (15)



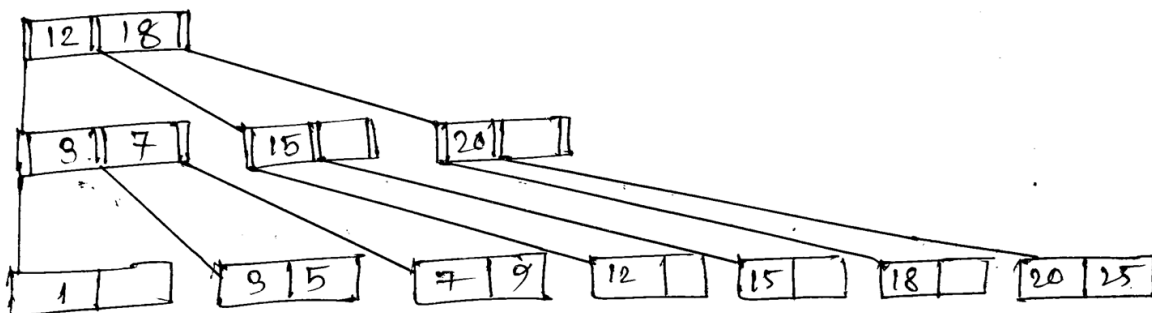
inserting (1)



inserting (18)

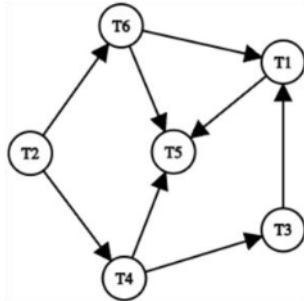


inserting (25)

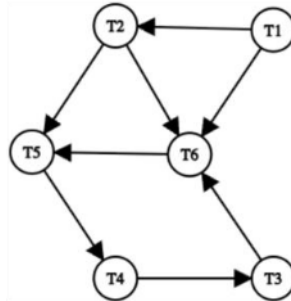


4. a) Explain the dirty read (uncommitted dependency) problem with an example in the context of transactions. 2  
+  
6  
=  
8
- b) Consider the following precedence graphs and determine whether the transactions are serializable. If they are serializable, find all the serial schedules; otherwise, explain why they are not serializable.

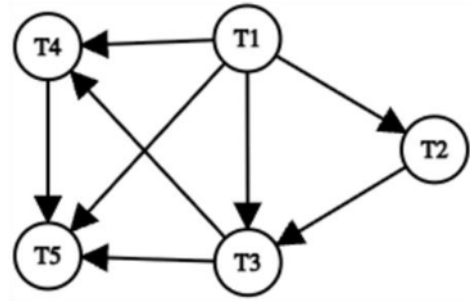
i.



ii.



iii.



#### 4) a)

A dirty read occurs when a transaction reads data that has been modified by another transaction that has not yet committed. If the other transaction is later rolled back, the data read by the first transaction becomes invalid or inconsistent.

- It's called "uncommitted dependency" because one transaction depends on data that is not yet committed by another transaction.
- This problem occurs in low isolation levels, especially Read Uncommitted.

Let's consider 2 transactions, T1 & T2:

| T1             | T2            |
|----------------|---------------|
| Write A := 500 |               |
|                | Read A := 500 |
| Rollback       |               |
|                | A := 500+100  |

In transaction T1, the value of A becomes 500. Before T1 commits, transaction T2 reads this value of 500. If T1 later rolls back due to a system failure or error, the change to A is discarded. However, T2 has already read 500 into memory and may perform operations using it, which are now meaningless because the value of A never actually exists in the database.